10.3.1 Reduction to tridiagonal form

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Week 10 due Mar 27, 2025 22:10 IST

Video

on here. The important thing is when we now apply

this from the right to this matrix, the first column of the matrix is not

affected so these zeros that we just introduced here remain zeros. The second

observation is that if A is a Hermitian matrix then this right here is also a

Hermitian matrix if these are Householder transformations. And what

does that mean? That means actually that

7:20 / 7:20

2.0x

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Reading assignment

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Read Unit 10.3.1 of the notes. [LINK]

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Why only tridiagonal (1:30)?

At around 1:30, Robert is asking us to think about why we can't just diagonal...

1

Homework 10.3.1.1

1/1 point (graded)

A more straight forward way of updating A_{22} is given by

$$\begin{array}{rcl} A_{22} & := & (I - \frac{1}{\tau} u_{21} u_{21}^H) A_{22} (I - \frac{1}{\tau} u_{21} u_{21}^H) \\ & = & \underbrace{(A_{22} - \frac{1}{\tau} u_{21} \underbrace{u_{21}^H A_{22}}_{y_{21}^H}) }_{B_{22}} (I - \frac{1}{\tau} u_{21} u_{21}^H) \\ & = & B_{22} - \frac{1}{\tau} \underbrace{B_{22} u_{21}}_{x_{21}} u_{21}^H . \end{array}$$

This suggests the steps

- Compute $y_{21} = A_{22} u_{21}$. (Symmetric matrix-vector multiplication).
- Compute $B_{22} = A_{22} - \frac{1}{\tau} u_{21} y_{21}^H$. (Rank-1 update yielding a *nonsymmetric* intermediate matrix).
- Compute $x_{21} = B_{22} u_{21}$. (Matrix-vector multiplication).
- Compute $A_{22} = B_{22} - \frac{1}{\tau} x_{21} y_{21}^H$. (Rank-1 update yielding a *symmetric* final matrix).

Estimate the cost of this alternative approach. What other disadvantage(s) does this approach have?

- [x] done

Solution

For a complete solution, see the notes .

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Answers are displayed within the problem

Homework 10.3.1.2

1/1 point (graded)

You may want to start by executing `git pull` to update your directory `Assignments`.

In directory `Assignments/Week10/matlab/`, you will find the following files:

- `Housev1.m`: An implementation the function Housev1 mentioned in the unit.
- `TriRed.m`: A code skelleton for a function that reduces a symmetric matrix to a tridiagonal matrix. Only the lower triangualr part of the input and output are stored.

  ```
  [ T, t ] = TriRed( A, t )
  ```

  returns the diagonal and first subdiagonal of the tridiagonal matrix in `T`, stores the Householder vectors below the first subdiagonal, and returns the scalars τ in vector `t`.

- `TriFromBi.m` : A function that takes the diagonal and first subdiagonal in the input matrix and returns the tridiagonal matrix that they define.

```
T = TriFromBi( A )
```

- `test_TriRed.m` : A script that tests `TriRed` .

With these resources, you are to complete `TriRed` by implementing the algorithm in Figure 10.3.1.2.

Be sure to look at the hint in the notes!

- [x] done

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